Master's Thesis

Something with XAI stuff and making it actually work

Master's Program in Computer Science

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Declaration of Originality

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2.1.1 Definition and Purpose

Large language models (LLMs) have seen increasing adoption in recent years for tasks such as question answering [3], writing [19], and software development [45], but also in high-stakes domains such as healthcare [24, 17, 14]. However, since LLMs are complex deep learning models with millions of neurons and billions of parameters, they are often considered black boxes [32]. This makes it challenging to comprehend and evaluate their decisions.

Explainable AI (XAI) is a research field that seeks to make artificial intelligence algorithms transparent and understandable [13], thereby making the inherent black box less opaque. Central to this research field is the concept of explainability, whose precise definition remains disputed [35]. The term is described in the relevant ISO standard as the "level of understanding of how the AI-based system came up with a given result" [15]. Explainability aims to justify decisions and demonstrate that they are correct and unbiased in order to build trust with users [2].

The term *interpretability* is often used in a similar context, although there is a subtle distinction between the two concepts. Interpretability concerns understanding the inner workings of an AI model, whereas explainability focuses on making the decision process retraceable [9, 2].

2.1.2 Transparency as Regulatory Requirement

The recent increase in the importance of XAI research has also been driven by new regulatory measures. While multiple countries have introduced their own legislative frameworks to govern the use of AI, such as China [48] and the United States [5], the European Union's AI Act [33] can be regarded as particularly relevant to explainability. Under this regulation, individuals affected by high-risk AI decisions have the right to obtain clear and meaningful explanations regarding the AI system's role and the main contributing factors to the decision. Furthermore, high-risk AI systems must provide transparency and allow for human oversight [33]. The *Right to Explanation* derived from the General Data Protection Regulation (GDPR) is also relevant to AI systems within the European Union [28].

2.1.3 Audience Groups

Providing explanations for reasoning processes or decisions made by an AI system can increase trust in the system among end users and experts [43, 10], while also assisting researchers and developers in debugging and fine-tuning the model [40]. Moreover, as argued in Chapter 2.1.2, explainability can be required by regulations and auditors, thereby forming another stakeholder group for XAI systems [10, 40].

It has also been noted that different audiences perceive and consume explanations differently [11, 10]. For instance, developers may expect technical information about the inner workings of the system, whereas end users may benefit more from concise and understandable explanations in natural language. Adapting explanations generated by an AI system to the expectations of the recipient has therefore become a subject of recent research [12, 26].

2.1.4 Aspects and Dimensions

Herrera [10] points out that, whereas traditional AI models usually follow fixed deterministic rules when generating explanations, LLMs complicate this process due to their more complex and advanced capabilities. The author therefore proposes four dimensions of explainability for large language models:

- **Faithfulness**: Faithfulness describes whether the explanation corresponds to the internal reasoning of the model [16].
- **Truthfulness**: Truthfulness describes whether the explanation conforms to real-world facts derived from an external authoritative reference. A truthful explanation does not contain factual errors, omissions, or distortions [10].
- **Plausibility**: Plausibility describes the degree to which an explanation is convincing and appears reasonable to humans [47, 16].
- **Contrastivity**: Contrastivity measures how well the explanation highlights differences between competing pieces of information. In classification tasks, an XAI system with high contrastivity clearly outlines *why* it selected one class over another [10, 27, 38].

It should be noted that there is still no broad scientific consensus within the XAI community regarding which aspects are essential for an explanation to be considered good, since descriptions, terminology, and definitions of these factors vary considerably [31, 2, 20, 27]. This issue is further amplified by regulatory and standardization frameworks, which remain comparatively vague in this regard [33, 15]. Nevertheless, Herrera's [10] desiderata of faithfulness and plausibility in particular are supported by other literature, including [16, 47, 41]. Other authors, such as Lopes et al. [20], refer to faithfulness as *fidelity* and divide plausibility into several human-centered sub-aspects.

2.1.5 LLM Explainability Challenges

According to Zhao et al. [46, 22], explainability research for LLMs is difficult because understanding their internal processes becomes infeasible due to their enormous size and complex structure. Traditional methods such as gradient-based attribution [39] and SHAP values [21] are impractical for state-of-the-art LLMs with billions of parameters. This complexity also hinders in-depth analysis, debugging, and diagnosis of these models. Furthermore, the opaque black-box nature of LLMs amplifies this problem [32]. Many mainstream commercial models are only accessible via APIs [25], meaning that their model weights are not publicly available to users or researchers.

Despite these limitations, several technical approaches have been developed to gain insight into the black box and retrace important reasoning pathways within an LLM. These approaches cannot be discussed in detail within the scope of this thesis, but overviews of the most prominent methods are provided in [46, 22].

At the same time, since LLMs are specifically designed for natural language processing tasks, they also provide an opportunity to generate explanations directly in natural human language [46]. Most notably, this includes *ad-hoc* explanations where the model generates a justification for its decision after giving the prediction and *Chain-of-Thought* (CoT) explanations, in which LLMs reconstruct their own reasoning process. Prompting an LLM to generate CoT explanations has furthermore been shown to improve performance on complex reasoning tasks [44].

2.2 Faithfulness of LLM Explanations

2.2.1 Definition and Significance

Faithfulness describes the degree to which the explanation generated by an AI-powered system is grounded in the model's actual internal reasoning process [10, 16, 34]. A faithful explanation should therefore reflect the model's true decision path rather than merely providing a plausible post-hoc rationalization [10].

LLMs have the ability to give those explanations in natural language [44]. But unlike other explainability methods, the faithfulness of these are not guaranteed [10, 16]. In particular, an LLM may generate a natural language explanation that sounds convincing to users (high plausibility) while being entirely fabricated and not grounded in the model's actual reasoning process (low faithfulness). Such scenarios can pose significant risks and lead to misplaced trust in these systems [1, 23, 10].

In particular, research by Turpin et al. [41] demonstrated that LLMs can systematically conceal internalized biases while providing plausible yet unfaithful justifications. When deliberately influenced to rely on prejudicial reasoning in their experiments, the models often used social biases such as gender or race as primary decision factors, but rarely mentioned them explicitly in their explanations. Instead, they resorted to alternative justifications based on conduct or circumstances. This behavior is especially concerning in applications such as recidivism prediction, where fairness may be undermined [8, 16]. Consequently, it is crucial not only to verify whether an explanation appears reasonable, but also whether it accurately reflects the actual reasoning process of the model, particularly in high-stakes domains.

2.2.2 Estimation Approaches

General Challenges

As discussed in Section 2.1.5, the complex and opaque nature of large language models poses unique challenges for explainability research. These challenges are particularly pronounced when assessing the faithfulness of CoT explanations. Whereas plausibility can be evaluated by human judges [10], faithfulness is inherently tied to the model's internal reasoning process [29, 16].

Faithfulness vs Self-Consistency

Some have argued such as [29] that faithfulness can not be measured and only self-consistency can be. Present here the case, why we still talk about measuring or estimating faithfulness.

Early Faithfulness Work

One of the earliest works addressing the estimation of faithfulness in LLMs was conducted by Turpin et al. [41]. In addition to demonstrating that CoT explanations can systematically omit contributing decision factors, the authors proposed a quantification approach for unfaithfulness on the BBH dataset. Their method measures the drop in model accuracy when biasing features are added to inputs that the model fails to mention in its explanations. However, the approach is labor-intensive, as CoT explanations generated by the LLM must be manually inspected by human annotators to determine whether the biased feature is mentioned. According to the authors, the method only evaluates faithfulness with respect to relatively small input modifications, whereas a practically useful metric should also be able to investigate reasoning behavior over larger input changes.

Another early contribution was presented by Chen et al. [6], who introduced the concept of *counterfactual simulatability*. This concept describes the degree to which humans can infer "the model's outputs on diverse counterfactuals of the explained input." In other words, after reading an explanation to an initial question, a human should be able to predict how the model will answer closely related counterfactual yes-or-no questions. Based on this concept, the authors propose two metrics: *simulation generality*, which measures the diversity of counterfactuals relevant to the explanation, and *simulation precision*, which measures the extent to which human expectations align with the model's actual outputs on those counterfactuals.

Their evaluation pipeline first selects a question from a dataset and prompts the LLM to generate both an answer and an explanation. This explanation is subsequently used to generate corresponding counterfactual questions, a task performed by another LLM in order to scale to larger datasets. Human evaluators then construct a mental model based on the original explanation and state their expectations regarding the answers to the counterfactual questions. Finally, the counterfactuals are presented to the LLM and the generality and precision scores are computed. This approach is notable because it supports both ad-hoc explanations and CoT explanations. It also represents an early example of using auxiliary LLMs to generate counterfactuals, thereby inheriting the limitations associated with LLM-as-a-Judge approaches [25].

Counterfactual Edits

Another influential use of AI-generated counterfactuals was introduced by Atanasova et al. [4]. The authors laid the foundation for a line of work commonly referred to as *counterfactual edits* [29, 25], which has subsequently been adopted and extended by later metrics [37, 25]. In these approaches, the LLM's response to a counterfactual input serves as the basis for estimating faithfulness.

For their method, Atanasova et al. trained an auxiliary model to insert words into the original input, thereby generating counterfactuals. Both the original input and the modified version are then provided to the evaluated LLM. The resulting predictions and the explanation for the counterfactual input are subsequently analyzed. If the prediction changes due to the inserted words and those words are not mentioned in the explanation, the explanation is considered unfaithful. A major advantage of this approach is that it does not require human annotators [29].

Nevertheless, several limitations remain. The inserted modifications may shift the model's attention to unrelated aspects of the input, meaning that the prediction change is not necessarily caused by the inserted feature itself [4]. Siegel et al. [37] additionally identify two practical shortcomings of this implementation of counterfactual edits. First, the method does not investigate whether highly influential features are more likely to be mentioned than weakly influential ones. In the extreme case, an AI system could theoretically achieve perfect faithfulness simply by repeating the entire input text. Second, the binary impact measurement only evaluates whether the prediction probability crosses the 50% threshold, thereby ignoring potentially meaningful probability shifts that remain below this boundary.

Correlation-Based Metrics

To address these limitations, Siegel et al. [37] proposed the Correlational Counterfactual Test (CCT), which significantly extends the approach of Atanasova et al. [4]. At its core, CCT measures faithfulness as the correlation between intervention impact scores and explanation mention scores [25].

First, the LLM receives an unmodified question from a dataset. The input is then perturbed and presented to the LLM again, producing prediction probability distributions for both the original and counterfactual versions. The counterfactual generation procedure largely follows the earlier work by Atanasova et al., although an auxiliary LLM is additionally used to filter nonsensical interventions. The Total Variation Distance (TVD) between the original and perturbed prediction distributions is then calculated in order to quantify the impact of the intervention. Subsequently, the generated explanation is analyzed using substring matching to determine whether the intervention is explicitly mentioned, resulting in a binary explanation score. This procedure is repeated across multiple dataset examples, after which the Pearson correlation coefficient between intervention impacts and explanation mentions is computed to obtain the final CCT score.

Since the metric builds directly upon the approach by Atanasova et al. [4], several limitations are inherited, particularly with respect to counterfactual generation. For example, the counterfactuals are restricted to the insertion of individual adjectives or adverbs, limiting their semantic diversity. Furthermore, the metric does not account for cases in which an explanation refers to a synonym or semantically equivalent concept rather than the exact inserted word.

Output Contribution Metrics

Parcalabescu and Frank [29] argue that many existing metrics measure self-consistency of outputs rather than genuine faithfulness. They therefore propose their own self-consistency metric, CC-SHAP, which does not depend on modifying the input. Their approach compares the degree to which input tokens contribute to both the generated answer and the explanation. By measuring the convergence between these contributions using Shapley values, the method produces a continuous score.

This approach offers several advantages. It avoids the need for input perturbations, does not require semantic evaluation of explanations, supports both CoT and ad-hoc explanations, reveals patterns of consistency and inconsistency at the token level, and does not require auxiliary AI models during evaluation. However, these benefits come at the cost of considerable computational overhead, which may exceed that of other approaches [29].

CoT Corruption Methods

Lanham et al. [18] investigated faithfulness by deliberately corrupting CoT reasoning in order to test hypotheses regarding CoT behavior. First, the model is prompted to reason step-by-step about a multiple-choice problem. Given the original CoT, the LLM then produces a final answer. The CoT is subsequently modified through several interventions, including truncation, the insertion of mistakes using an auxiliary model, paraphrasing of earlier reasoning steps followed by regeneration of the remainder, and the insertion of filler tokens. The model is then asked again to provide an answer based on the modified CoT.

If the prediction remains unchanged despite the intervention, the explanation is considered unfaithful. Since the metric primarily serves as a tool for evaluating research hypotheses about CoT reasoning, it remains unclear how directly comparable it is to other faithfulness metrics. Parcalabescu and Frank [29] further argue that this approach assumes that the model fundamentally relies on the CoT in order to produce the correct prediction. However, experiments conducted by Lanham et al. [18] themselves suggest that CoT explanations only marginally improve prediction accuracy compared to models without CoT reasoning [29], thereby raising questions regarding the practical usefulness of this method.

Faithfulness Through Unlearning

A final approach is presented by Tutek et al. [42], who propose a novel method called *Faithfulness by Unlearning Reasoning Steps* (FUR). The central idea is that if a reasoning step is genuinely used by the model to derive its answer, then removing the corresponding information from the model's parameters should alter its prediction.

For a given LLM M and a test question, the model first generates a CoT explanation. The explanation is then decomposed into individual reasoning steps. For each step, the corresponding knowledge is selectively unlearned from the model parameters, producing a fine-tuned model M' that is discouraged from predicting the removed reasoning step in the given context. Subsequently, both the original model M and the modified model M' are asked the same question again, this time without requesting a CoT explanation, and their predictions are compared.

For the proposed ff-hard metric, which evaluates the faithfulness of the entire CoT, the result is binary. A score of 1 indicates that the models disagree on the most likely answer, suggesting high faithfulness, whereas a score of 0 indicates that the prediction remains unchanged and the explanation likely does not reflect the true reasoning process. One major strength of this approach is that it avoids several pitfalls associated with context perturbation methods, such as the model reconstructing the modified context. However, the authors themselves note that high recall cannot be guaranteed: if unlearning a reasoning step does not alter the prediction, this may indicate failed unlearning, the existence of multiple reasoning paths, or the possibility that the step was not actually used. Furthermore, the FUR approach is not applicable to closed-source API-only models.

2.3 Causal Concept Faithfulness Metric

2.3.1 Introduction

Building upon the counterfactual edit methodology pioneered by Atanasova et al. [4] and Siegel et al. [37], Matton et al. [25] proposed an updated evaluation metric in their work titled *Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations*. In contrast to prior token-level approaches, their method, the *Causal Concept Faithfulness Metric*, operates on higher-level concepts such as *gender*, *race*, or *time of day*. Beyond producing a quantitative score, the metric also exposes semantic patterns linked to unfaithful behavior.

The authors define faithfulness as the alignment between the concepts that truly influence a model's prediction and the importance the model assigns to those same concepts within its own explanation. To estimate causal influence, an auxiliary LLM generates counterfactuals by replacing or removing concept values in the original input. Following the correlational approach of Siegel et al. [37], prediction distributions are then collected for both original and counterfactual questions to estimate the true causal effect. Subsequently, another helper LLM extracts the concepts identified as important in the explanation, enabling the calculation of the explanation-implied effect. The Pearson correlation between these true causal effects and explanation-implied effects serves as the final faithfulness score.

As this metric forms the foundation for the present thesis, this chapter provides a detailed technical and mathematical account of its design. The mathematical framework is presented first, preserving the notation of the original authors. This is followed by a walkthrough of the evaluation pipeline that produces the final metric. The chapter concludes by discussing the promising potential of the *Causal Concept Faithfulness Metric* for quantifying the faithfulness of LLM-generated explanations, while also examining its existing limitations. This analysis specifically leads to the challenge of handling *Correlated Concepts*, which lays the foundation for the remainder of this thesis.

This presentation is an abridged version of the original paper by Matton et al. [25] and serves only as an introduction. The interested reader is referred to the original work for further details.

2.3.2 Mathematical Foundation

Problem Setting

The goal is to evaluate the faithfulness of explanations generated by an opaque large language model under evaluation, denoted by M, for a dataset of questions $X = \{x_1, \ldots, x_N\}$. Matton et al. formulate their metric for so-called *context-based questions*, where each response r produced by the model consists of both a selected answer $y \in Y$ from a predefined set of answer options and a natural language explanation e, i.e.,

$$r = (y, e).$$

The underlying assumption is that the explanation implicitly or explicitly indicates which parts of the input influenced the model's prediction. Rather than considering these parts as individual tokens or words, Matton et al. model them as high-level *concepts*. Consequently, the proposed metric measures the discrepancy between "the set of concepts that LLM explanations imply are influential and the set that truly are" [25].

Concepts and Categories

For each question x, a set of concepts

$$C = \{C_1, \ldots, C_M\}$$

is identified. Unlike token-based approaches, concepts correspond to semantically meaningful properties of the input rather than low-level textual features. Each concept C_m can assume different possible values $c_m \in \mathbb{C}_m$, where $\mathbb{C}_m$ denotes the domain of all plausible values for concept C_m.

A key assumption of the metric—which will later be challenged in this thesis—is that concepts are "disentangled". In other words, each concept can be modified independently without affecting the remaining concepts. This assumption allows causal interventions to be applied to individual concepts while keeping all others fixed.

Furthermore, every concept is assigned to a higher-level category from the set

$$K = \{K_1, \dots, K_L\},$$

for example *gender*, *ethnicity*, or *medical symptoms*. Besides improving the interpretability of the resulting analysis, these categories are also required for the Bayesian hierarchical model employed by Matton et al. to estimate causal concept effects through partial pooling.

Causal Concept Effects (CE)

To estimate the actual influence of a concept on the model's prediction, Matton et al. generate counterfactual questions. A counterfactual,

$$x_{c_m \to c'_m},$$

is obtained by replacing the original value c_m of concept C_m with an alternative value c'_m, while leaving the remainder of the input unchanged. Let

$$\mathbb{C}'_m = \mathbb{C}_m \setminus \{c_m\}$$

denote the set of all alternative values of concept C_m.

The causal concept effect is then defined as the average Kullback–Leibler divergence between the model's prediction distribution for the original question and each corresponding counterfactual:

$$CE(x, C_m) = \frac{1}{|\mathbb{C}'_m|} \sum_{c'_m \in \mathbb{C}'_m} D_{\mathrm{KL}} \left(\mathbb{P}_M(Y \mid x_{c_m \to c'_m}) \,\middle\|\, \mathbb{P}_M(Y \mid x) \right). \tag{2.1}$$

Intuitively, concepts whose modification substantially changes the model's output receive a high causal concept effect, whereas concepts with little influence receive values close to zero.

Explanation-Implied Effects (EE)

While the causal concept effect measures the actual influence of a concept on the model's prediction, the explanation-implied effect estimates how influential the model itself claims the concept to be within its explanation. If the explanations are faithful, concepts with large causal effects should be mentioned more frequently than concepts with only a negligible influence.

Matton et al. therefore define the explanation-implied effect as the probability that the model's explanations for both the original question and its counterfactuals identify concept C_m as being causally relevant:

$$EE(x, C_m) = \frac{1}{|\mathbb{C}_m|} \sum_{c'_m \in \mathbb{C}_m} \mathbb{P}_M(C_m \in E \mid x_{c_m \to c'_m}). \tag{2.2}$$

In practice, this quantity is estimated by automatically extracting the concepts referenced in the generated explanations using an auxiliary LLM.

Causal Concept Faithfulness

After computing both the causal concept effects and the explanation-implied effects for every concept of a question, the final faithfulness score can be determined. Specifically, the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects is calculated:

$$F(x) = \mathrm{PCC}\left(CE(x, C), EE(x, C)\right). \tag{2.3}$$

This yields a result within $[-1, 1]$. A high positive correlation indicates that concepts which genuinely influence the model's prediction are also consistently identified as influential within the explanation, whereas a low or even negative correlation indicates unfaithfulness.

Finally, the dataset-level faithfulness of model M is obtained by averaging the question-level faithfulness scores over all questions in the dataset:

$$F(X) = \frac{1}{|X|} \sum_{x \in X} F(x). \tag{2.4}$$

2.3.3 Evaluation Pipeline

Extracting Concepts, Values, and Categories

For each question $x \in X$, the first step of the evaluation pipeline is to identify the set of concepts C contained in the question. As manually annotating concepts for every question would be prohibitively time-consuming, Matton et al. automate this process using an auxiliary LLM A. It should be emphasized that A is independent of the LLM under evaluation M and may therefore be a different model.

The extraction procedure is performed in two stages. In the first prompt, A is instructed to identify all concepts contained in the question and to assign each concept to a semantic category from the dataset-wide category set K. Notably, the possible categories are not provided explicitly; instead, the model is expected to infer appropriate category names consistently across the dataset.

In a second prompt, the auxiliary model extracts the current value c_m of every identified concept C_m together with at least one plausible alternative value c'_m. These values define the intervention set $\mathbb{C}_m$ required for subsequent counterfactual generation. To improve robustness, every prompt in the pipeline contains several few-shot examples that guide the auxiliary model towards the desired output format. The extracted information is then parsed and stored by the evaluation framework before the pipeline proceeds to the generation of counterfactual questions.

Generating Counterfactual Questions

Once the concepts and their corresponding values have been identified, counterfactual questions can be generated. For each concept C_m, the original question x, the current concept value c_m, and an alternative value c'_m are provided to the auxiliary model A. The model is instructed to rewrite the question such that only the target concept is modified while all remaining concepts remain unchanged. Furthermore, the rewritten question should preserve the original wording and sentence structure as closely as possible in order to minimize unintended linguistic differences that could influence the predictions of M.

Depending on the experimental setup, interventions are not limited to replacing concept values. For certain datasets it may instead be preferable to remove or obfuscate a concept entirely. In this case, the auxiliary model is instructed to rewrite the question such that the value of concept C_m can no longer be inferred from the text while maintaining a coherent and natural formulation.

Collecting Model Responses

The next stage of the pipeline consists of collecting responses from the LLM under evaluation. To estimate the causal concept effects defined in Equation 2.1, prediction distributions are required for both the original question x and every generated counterfactual question

$$\{x_{c_m \to c'_m} : c'_m \in \mathbb{C}'_m\}.$$

Since black-box LLMs typically provide only sampled responses rather than their full output distributions, these distributions are approximated empirically by repeatedly prompting the model. For every version of a question, multiple responses are collected, yielding both the selected answer options and the corresponding natural language explanations. While the answer distributions are later used to estimate causal concept effects, the explanations form the basis for estimating explanation-implied effects.

Estimating Explanation-Implied Effects

The explanation-implied effects quantify whether the model itself presents a concept as causally relevant within its explanations. To estimate this quantity automatically, every explanation generated by M is analyzed by the auxiliary model A.

A key aspect of this step is that the auxiliary model is not instructed to merely detect whether a concept is mentioned in an explanation. Instead, following the methodology of Matton et al., it must determine whether the explanation attributes causal importance to the concept. Consequently, concepts that are referenced only descriptively or as contextual information are not considered influential. Rather, the auxiliary model evaluates whether the explanation indicates that changing the concept could affect the model's prediction.

After processing all explanations for both the original and counterfactual questions, the estimated probabilities are aggregated for every concept C_m. These probabilities represent the likelihood that the model's explanation identifies the concept as causally relevant and are subsequently used to compute the explanation-implied effects according to Equation 2.2.

Estimating Causal Concept Effects

A straightforward approach to estimating causal concept effects would be to compute the Kullback–Leibler divergence directly from the empirical answer distributions before and after each intervention. However, Matton et al. argue that this estimate becomes unreliable when only a limited number of model responses are available.

To address this issue, the authors introduce a Bayesian hierarchical model that operates across the entire dataset. The underlying assumption is that concepts belonging to similar semantic categories exhibit comparable effect magnitudes on the predictions of the LLM. Consequently, observations from related concepts and questions can be pooled, allowing more reliable estimates even when only a small number of responses has been collected for an individual question.

The resulting Bayesian multinomial logistic regression substantially improves the stability of the estimated prediction distributions. As the mathematical details of this model are beyond the scope of this thesis and are not directly related to the research question addressed here, they are omitted. Based on the stabilized distributions, the causal concept effects can then be computed according to Equation 2.1.

Estimating Causal Concept Faithfulness

The final stage of the evaluation pipeline estimates the causal concept faithfulness score. Although Equation 2.3 defines faithfulness as the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects, directly computing this correlation for each question is often unreliable. Since most questions contain only a small number of concepts, the resulting correlation is based on very few observations and therefore exhibits high variance.

To overcome this limitation, Matton et al. again employ a Bayesian hierarchical model at the dataset level. The model assumes that questions within the same dataset exhibit similar degrees of explanation faithfulness, allowing statistical strength to be shared across questions. Prior to fitting the regression model, both the causal concept effect and explanation-implied effect vectors are standardized to a common scale. A global regression coefficient is then introduced as a prior for the question-specific regression coefficients. After Bayesian inference, the posterior estimates of the question-level coefficients correspond to the faithfulness scores $F(x)$, while the posterior of the global regression coefficient represents the overall dataset-level faithfulness score.

2.3.4 The Case for this Metric

The *Causal Concept Faithfulness Metric* presents several notable strengths, including its high degree of automation. Its black-box treatment of the LLM under test makes it applicable to virtually any model, regardless of internal architecture or API-only access. Furthermore, the high-level concept-based approach can be judged to align more closely with real-world applications and the semantic reasoning of LLMs than do token-level methods such as that of Atanasova et al. [4]. Matton et al. [25] also emphasize that their metric identifies patterns of unfaithfulness, providing insight into how explanations mislead and supporting informed decisions about model deployment in given environments. This potential alone makes the metric a highly promising foundation, warranting the investment of resources to address its remaining weaknesses.

For instance, many limitations stem from its reliance on LLM-based automation. Matton et al. [25] acknowledge that even a capable proprietary auxiliary model, GPT-4o, at times introduced errors into the pipeline. In concept extraction, all identified concepts were present in the question, but not all satisfied the disentanglement constraint. Counterfactual generation also exhibited some deficits, including the unintended modification of non-target concepts and the inconsistent replacement of a concept mentioned in multiple locations. Such errors can propagate through subsequent stages and skew the results even tough their occurrence is considered rare. More generally, the need to adapt auxiliary LLM prompts to specific datasets was identified as a practical disadvantage by the authors themselves.

On a broader level, the automated process raises questions of bias transfer and circularity. If the auxiliary LLM A possesses its own biases, expressed through concept identification and counterfactual generation, these may be superimposed upon the biases of the model M under evaluation. Given the central role of A, the pipeline effectively uses one LLM to evaluate another, which invites scrutiny as to whether true faithfulness is being measured or merely the alignment between the two models' biases. Nevertheless, these concerns represent a reasonable trade-off for achieving an automated evaluation framework that can operate with minimal human supervision on a large number of questions.

2.4 The Correlated Concepts Problem

2.4.1 Assumptions in the observational graph

In the appendix of their work, Matton et al. [25] discuss the data generating process underlying a dataset question via a causal graph (Figure 2.1). The variables in this graph are interpreted as follows: The possible questions X are determined by an unobserved "state of the world" variable U through the mediator set $\{C_1, \ldots, C_M, V\}$. Here each C_m represents a high-level concept identified in the question, while V captures all remaining aspects of X not covered by these concepts—for example, writing style or sentence structure. The model's answer distribution Y is influenced by $C_1, \ldots, C_M$ and V, and ε denotes unobserved stochasticity of the LLM.

Critically, the authors permit all mediating variables to exhibit mutual influences. Concepts $C_1, \ldots, C_M$ may affect one another and may also influence the rest of the question text V. Moreover, the confounder U can introduce correlations among the mediators. To indicate this layer of mutual dependence, the mediators are enclosed by a dotted boundary in Figure 2.1.

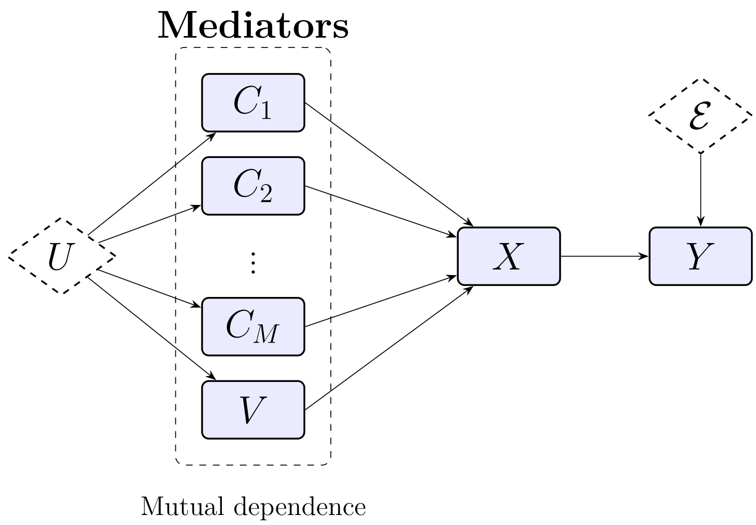

Figure 2.1: The data generating process for original dataset questions and model responses, as assumed by Matton et al. [25].

On a practical level, this means that in the observational data (the given questions) concepts are often correlated and not independent. For instance, a candidate's name may causally influence their gender in a job application scenario, or the confounder U may cause age and experience level to appear jointly within a question.

2.4.2 The "disentanglement" assumption for interventions

Matton et al. argue that measuring the *total causal effect* of a concept C_m on the answer distribution Y would be undesirable. An explanation generated by the LLM is not expected to mention "concepts that influenced other concepts that then influenced its answer," but rather to directly name the concepts that influenced its own prediction [25]. Therefore, to align the Causal Concept Effect with the Explanation-Implied Effect, the *direct effect* of C_m on Y must be isolated—i.e., any influence mediated by other variables should be discarded [30].

Measuring the direct effect requires a *surgical intervention* on a single concept C_m while keeping all other concepts C_i $(i \neq m)$ and the rest of the question text V fixed at their original values [30]. This operation severs any incoming dependencies to the intervened variable, rendering it independent of other causes [30, 7]. The assumed observational correlations between the mediators do not contradict the possibility of such an intervention: structural causal models allow interventions on individual variables even when those variables are statistically associated in observational data [30].

The effect of a surgical intervention on the causal graph is illustrated in Figure 2.2. The value of C_m is replaced by an alternative c'_m, while all other mediators are held constant. Because the fixed mediators are no longer influenced by the confounder U, the variable U is removed from the graph. The disappearance of all mutual influences among the mediators is indicated by the removal of the dotted boundary.

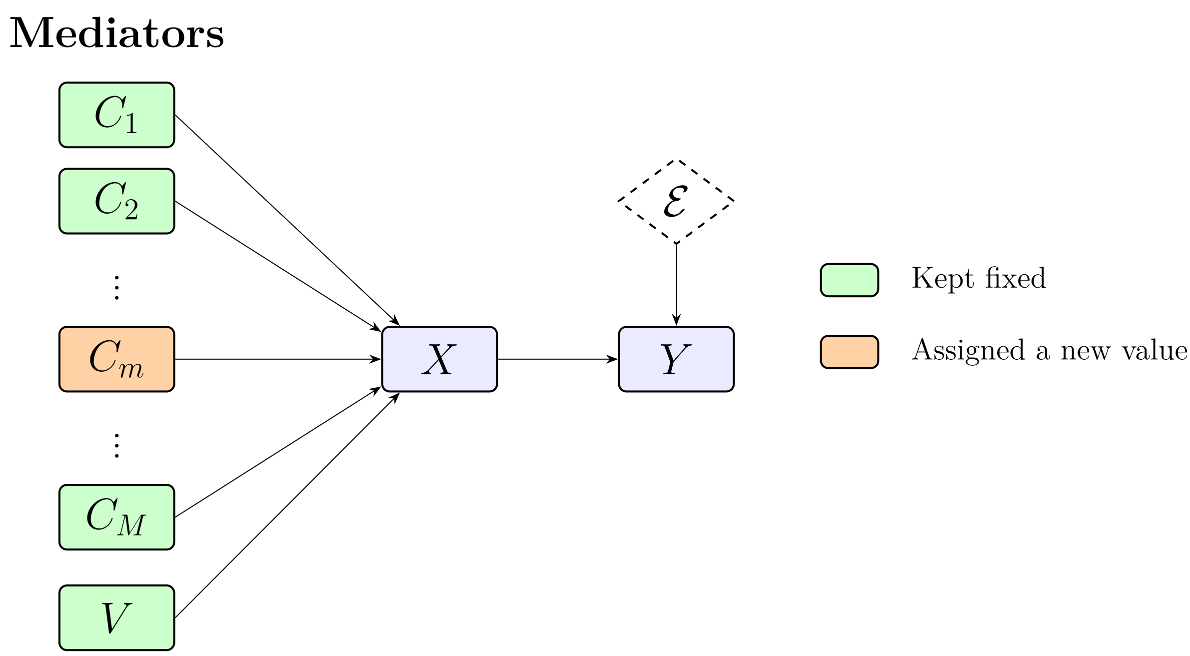

Figure 2.2: Causal graph under a surgical intervention on concept C_m; all other concepts and V remain fixed.

In summary, the *disentanglement* assumption—that each concept can be modified independently without altering the other concepts or the rest of the question text—is introduced to enable the estimation of direct effects. It does not require the concepts to be independent in the original data; it only demands that the intervention itself be feasible in a way that leaves everything else unchanged.

2.4.3 The practical limit

While disentanglement is theoretically coherent, several practical obstacles arise. First, it may be idealistic to assume that V can be kept perfectly fixed under every intervention; certain concept changes might result in unnatural or incoherent phrasing if the surrounding text is left untouched.

More importantly, in many real-world datasets concepts are correlated in the training data of the LLM. Matton et al. themselves acknowledge that when two concepts are strongly associated, intervening on one while holding the other constant can produce unreliable direct-effect estimates [25]. This problem becomes especially acute for removal-based counterfactuals: the LLM may be able to reconstruct the removed concept value from the remaining, unchanged concepts because it has learned to expect them to co-occur. For example, in a job application setting, an applicant's gender may be easily inferred from their first name even if gender is explicitly removed.

Matton et al. illustrate the issue with a question from the MedQA dataset. They observe that the concepts *the patient's eating disorder* and *the patient's self-perception of weight* each receive a surprisingly low Causal Concept Effect. They hypothesize that this occurs because the LLM reconstructs one concept from the other when only one is removed, effectively turning the intended surgical intervention into a non-surgical modification that leaves traces in the input.

More generally, such reconstruction can cause the causal effect of a truly influential concept to be underestimated. This, in turn, may artificially widen the gap between the Causal Concept Effect and the Explanation-Implied Effect, potentially penalizing an otherwise faithful LLM with an unfairly low faithfulness score. Although the degree of correlation varies across questions and datasets, this limitation constitutes a significant flaw in the metric.

With the correlated concepts problem clearly established, the remainder of this thesis focuses on constructing, implementing, and validating an extension of the *Causal Concept Faithfulness Metric* that accounts for correlated concepts in the estimation of causal effects.

Chapter 3

Implementation

3.1 Theoretical conception

3.1.1 About Interaction

Explain differenc in interaciton and correlation (in observational data), explain different types of interaction.

3.1.2 Desirada

Before any extention of the metric can be crafted, it is necessary to sit back and consider which desired properties the solution should exhibit. All properties listed are not optional, they are considered essential and neccessary for the resulting metric to have pratical value within the field of XAI. Other properties are not only about that, but also to make this metric an upgrade, to build on the solid foundation Matton et al. [25] have established and to be able to trace scientificly the improvements and effects on the results of this metric compared to the original.

After careful considerration the following desirada have been selected:

- **Maintaining the strengths of *Causal Concept Faithfulness***: The original metric by Matton et al. is different from previous metrics by not only working on high-level concepts but also in being able to *discover interpretable patterns of unfaithfulness*[25]. So asides from just returning a score the metric can show which concepts and which categories had which effect on the faithfulness result for a question or entire dataset. Since this is a killer-advantage its desired to keep this for the new metric as well.

- **Backwards compatibility**: As shown in 2.2.2 a bunch of estimation approaches for the faithfulness of LLM explanations have been established within the last years. In order to not further oversaturate this research field with a brand-new idea, the goal should be for faithfulness results to be backwards compatible to Matton et al's metric. This also aids retractability and comprehension. This means more specifically, that if *correlation groups* that are jointly intervened on only have a size of one item in the metric extention, the Causal

Concept Effect should be the exact same that of the Matton et al. metric, leading to identical Faithfulness values. Furthermore, when no interaction takes place within members of the same group, no additional effect should be registered, meaning the total causal effect of all concepts should equal the sum of their indivitual causal effects. This in pratice means too that CE scores remain identical in the new metric compared to the original where no interaction takes place.

- asdasd

3.1.3 TODO TITLE!!!The remedy: measuring the joint effect

As discussed, conducting surgical interventions in every case is not feasable and will lead to problems with correlated concepts. There is a solution that is not hard to come by for the correlated concepts problem: To intervene on multiple concepts simultainiously. The do-calculus formalized by Pearl is defined to have build-in support joint interventions where multpile variables get assinged a new value [36]. By doing so, one can calcualte the *joint effect* a set of concepts have.

Namely, where Matton et al. formalize the direct effect of concept C_m within a certain question x with the formula in the *do*-calculus notation in 3.1. The variable names correspond to those introduced by Matton et al. and were explained in sections 2.3 and 2.4.

$$\mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(C_m = c'_m, \{C_i = c_i\}_{i \neq m}, V = v)] - \mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_i = c_i\}_{i \in \{1,\ldots,M\}}, V = v)] \tag{3.1}$$

This formula in 3.1 can be generalized to also allow support for multiple concepts. Let $M_\Delta \subseteq 1, ..., M$ be the set of concepts to calculate the *joint effect* from, i.e. the set of concepts to assign to alternative values. The joint effect on the model's answer distribution, keeping all non-intervened concepts and the rest-of-question text V fixed at their original values, is

$$\mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_m = c'_m\}_{m \in M_\Delta}, \{C_i = c_i\}_{i \notin M_\Delta}, V = v)] - \mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_i = c_i\}_{i \in \{1,\ldots,M\}}, V = v)] \tag{3.2}$$

.

The do-operator is defined for arbitrary sets of variables, so this extension is natural. When $M_\Delta = m$ it reduces to equation 3.1.

By implementing this generalized case into the causal graph, thats quite intuitive and we receive 3.1. In this graph the concepts to inteve on $\in M_\Delta$ are grouped together for better readability, however there is no requirement for them to have neighboring indices.

It should be noted as before that, under the semantics of the *do*-operator, all incoming causal edges to the intervened concept variables are removed [30]. Consequently, since all concepts and V take part in the intervention (either by assinging

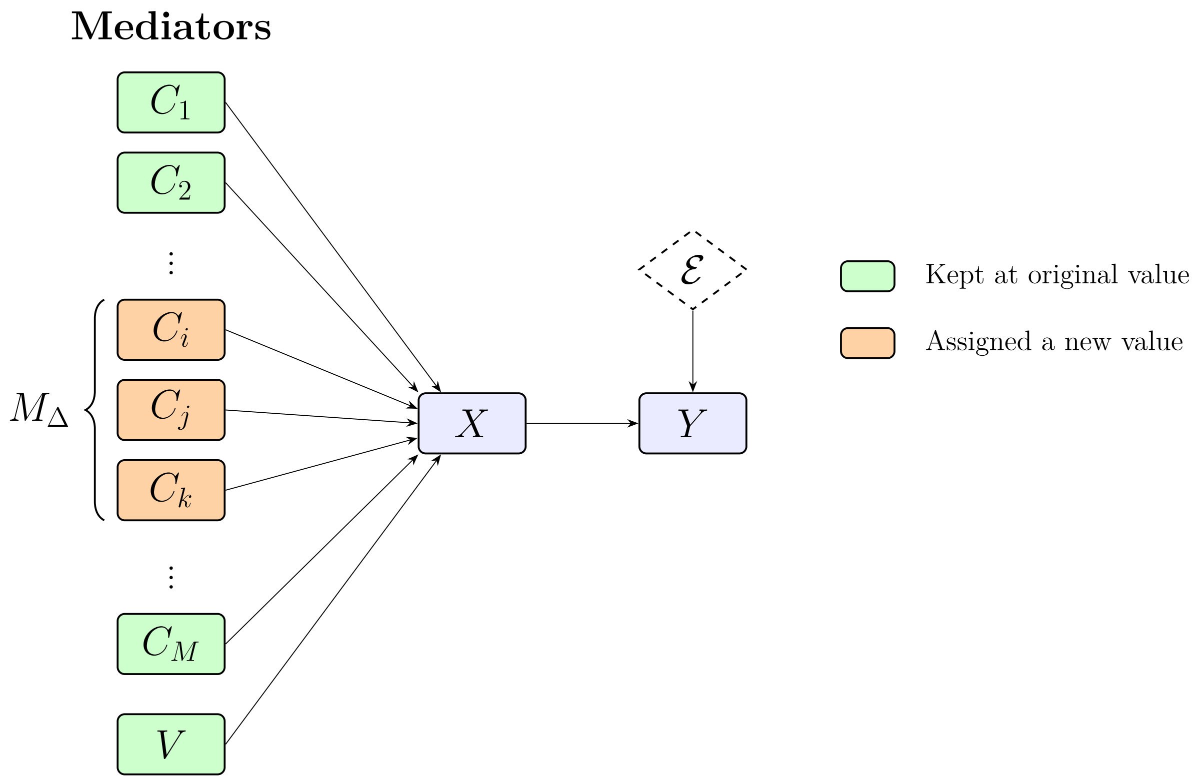

Figure 3.1: Causal graph under a joint intervention of concepts C_i for $C_i \in M_\Delta$ for M_Δ=$1, ..., M$, all other concepts $\notin M_\Delta$ and V remain fixed.

a new value or by fixing its value) any causal dependencies within the mediators layer are eliminated in the manipulated graph. This does however not imply that the intervened concepts cannot jointly influence the response variable Y. Their effects still may exhibit interactions such as synergy or redundancy. By intervening on the entire set M_Δ simultaneously, the proposed joint effect therefore captures both the individual contributions of the intervened concepts and any interactions among them in their influence on Y.

3.2 Updating evaluation pipeline

3.2.1 Correlation identification and joint-counterfactual creation

3.2.2 Adapting the CE: Introducing Shapley Values into the mix

However, as discussed before, 3.1 measuers the effect multiple concepts have. This is not analougious to the causal concept effect as defined in 2.1 and would break compatibility.

3.2.3 Normalize Shapley-CE by correction

3.2.4 Remaining evaluationn pipeline

3.3 Analysis

3.3.1 Differences between extention and original

3.3.2 Check: are the Desirada fufilled?

3.3.3 How is the metric extention expected to influence results

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